

NOVY URENGOY, Russian Federation

WMO: 233580

Lat: 66.0742N

Lon: 76.5006E

Elev: 203

StdP: 14.59

Time Zone: 5.00 (E05)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-47.0	-42.2	-52.6	0.2	-45.4	-48.9	0.3	-42.1	29.7	11.6	26.2	8.2	3.5	30	0.857

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.5	(c) 82.7	(d) 63.7	(e) 77.7	(f) 62.4	(g) 73.1	(h) 60.1	(i) 66.3	(j) 78.0	(k) 63.8	(l) 74.8	(m) 61.4	(n) 71.1	(o) 9.2	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.8	83.2	70.4	59.2	75.8	68.3	57.1	69.9	66.1	31.0	78.0	29.2	74.7	27.5	70.2	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	21.6	19.4	-48.9	86.4	6.8	4.2	-53.8	89.4	-57.8	91.9	-61.6	94.2	-66.6	97.2
			-47.4	68.3	5.4	3.0	-51.3	70.4	-54.5	72.2	-57.5	73.9	-61.5	76.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.4	-11.3	-10.5	4.4	15.5	28.8	52.0	60.2	51.8	42.2	24.5	2.8	-5.4		
	DBStd	28.02	17.40	16.33	15.07	13.87	10.42	10.48	8.74	6.44	7.75	11.44	16.50	17.73		
	HDD50	11053	1900	1695	1412	1036	663	101	12	54	258	789	1417	1716		
	HDD65	15969	2365	2115	1877	1486	1121	406	203	410	684	1254	1867	2181		
	CDD50	632	0	0	0	0	8	160	329	111	24	0	0	0		
	CDD65	72	0	0	0	0	0	15	55	2	0	0	0	0		
	CDH74	850	0	0	0	0	0	192	632	26	0	0	0	0		
	CDH80	248	0	0	0	0	0	43	204	1	0	0	0	0		
	WSAvg	9.7	9.4	9.4	10.3	11.1	10.8	9.8	8.9	8.3	9.1	10.3	9.2	9.6		
	PrecAvg	31.8	2.4	2.2	2.4	2.4	2.3	2.6	2.3	3.0	2.4	3.7	3.1	2.8		
(15) Precipitation	PrecMax	36.3	4.2	4.2	4.1	4.2	4.2	4.5	4.9	5.8	4.7	6.5	5.7	4.2		
	PrecMin	23.7	1.1	0.7	0.3	0.4	0.9	1.2	0.5	1.6	1.1	1.8	1.3	1.1		
	PrecStd	3.2	0.9	0.9	0.8	0.9	0.7	0.8	1.0	0.9	0.8	0.9	1.0	0.8		
	PrecStd	3.2	0.9	0.9	0.8	0.9	0.7	0.8	1.0	0.9	0.8	0.9	1.0	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.8	26.7	34.2	42.9	69.7	85.5	89.3	77.2	69.8	50.3	35.3	29.9		
		MCWB	25.8	26.1	32.2	36.6	56.9	63.6	67.4	62.2	57.4	46.9	34.2	29.3		
	2%	DB	21.5	22.8	31.5	39.0	57.4	79.2	85.7	72.0	62.8	44.8	33.2	26.4		
		MCWB	20.7	22.0	30.2	34.6	49.4	61.6	65.8	60.8	54.6	43.2	32.6	25.7		
	5%	DB	17.6	17.5	28.8	36.0	48.4	75.0	81.8	67.5	58.0	41.2	30.5	23.5		
		MCWB	16.5	16.6	27.7	33.2	42.8	59.7	63.8	57.8	52.4	39.8	29.9	22.8		
	10%	DB	12.6	13.0	25.5	34.1	41.4	70.0	76.6	63.2	53.9	37.9	26.5	19.5		
		MCWB	12.0	12.4	24.3	32.4	37.4	57.2	62.9	56.0	49.4	36.7	25.8	18.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.0	25.9	32.4	37.4	56.3	67.2	70.0	64.8	59.9	47.4	34.2	29.2		
		MCDB	26.8	26.2	33.8	40.8	68.1	78.0	83.2	72.2	67.0	49.3	34.5	29.5		
	2%	WB	20.7	21.9	30.3	35.1	50.0	64.4	67.8	61.9	55.9	43.4	32.4	25.6		
		MCDB	21.4	22.6	31.3	37.6	56.0	75.4	81.3	69.3	60.7	44.4	32.7	26.1		
	5%	WB	16.8	16.6	27.9	33.6	43.0	61.6	65.5	59.3	53.2	39.9	29.8	22.8		
		MCDB	17.6	17.3	29.0	35.4	48.1	71.9	77.8	66.0	58.6	40.9	30.4	23.5		
	10%	WB	12.4	12.4	24.4	32.5	37.9	58.7	63.3	56.7	50.2	36.6	25.7	18.8		
		MCDB	12.9	13.0	25.5	34.4	40.9	67.8	74.4	61.9	53.3	37.8	26.4	19.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	15.3	17.5	16.7	12.2	17.2	17.5	14.8	12.0	9.8	13.3	14.1		
		MCDBR	17.1	15.9	16.9	11.9	20.9	24.6	23.8	21.6	18.2	10.5	13.2	11.2		
		MCWBR	16.9	15.7	16.3	9.5	13.3	11.7	9.0	11.3	10.9	8.7	13.1	11.2		
	5% WB	MCDBR	17.2	16.3	17.3	12.5	20.4	22.6	22.2	19.0	17.6	10.1	13.0	11.8		
		MCWBR	17.2	16.1	16.9	11.0	13.5	11.9	10.3	11.3	11.1	8.8	12.9	11.8		
(40) Clear-Sky Solar Irradiance	taub		0.210	0.262	0.315	0.331	0.350	0.372	0.401	0.386	0.328	0.280	0.201	0.210		
	taud		1.875	2.029	2.018	2.074	2.197	2.392	2.308	2.319	2.442	2.197	1.904	1.884		
	Ebn at Noon		63	182	227	259	268	266	251	242	234	177	63	6		
	Edh at Noon		6	19	31	38	38	32	34	31	22	16	5	1		
(44) All-Sky Solar Radiation	RadAvg		25	213	732	1398	1720	1695	1570	1051	592	200	53	5		
	RadStd		2	21	52	119	111	155	189	130	79	23	7	0		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+2.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page